Taylor and Maclaurin Series

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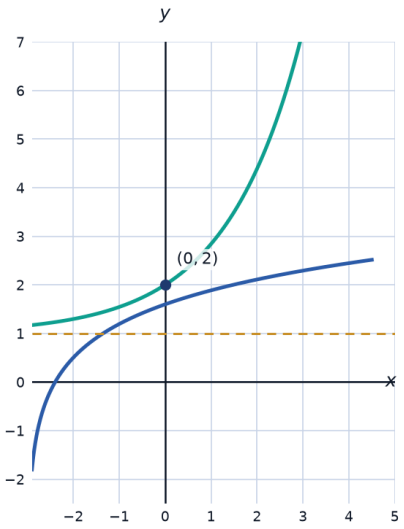
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: The Taylor series formula

State the general Taylor series formula for $f(x)$ centered at $x = a$.

2 [Foundational] • Target: The Taylor series formula

Find the Taylor series for $f(x) = \ln x$ centered at $a = 1$, up through the $(x - 1)^3$ term.



3 [Foundational] • Target: Common Maclaurin series

State the Maclaurin series for e^x, $\sin x$, and $\cos x$.

4 [Foundational] • Target: Common Maclaurin series

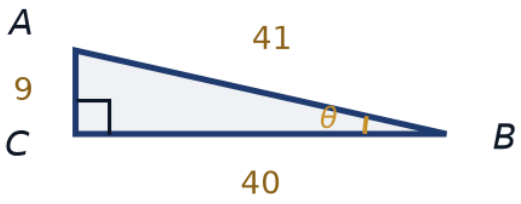
Use the Maclaurin series for e^x to find the Maclaurin series for e^{-x^2}.

5 [Foundational] • Target: Using known series to build new series

Use the Maclaurin series for $\sin x$ to find the first three nonzero terms of the Maclaurin series for $x \sin x$.

6 [Foundational] • Target: Using known series to build new series

Use the known Taylor series for $\cos x$ to estimate $\cos(0.1)$ using the first three nonzero terms.



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7 [Foundational] • Target: The Lagrange error bound

State the Lagrange Error Bound formula for the remainder $R_n(x)$ of a Taylor polynomial of degree n.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The Lagrange error bound

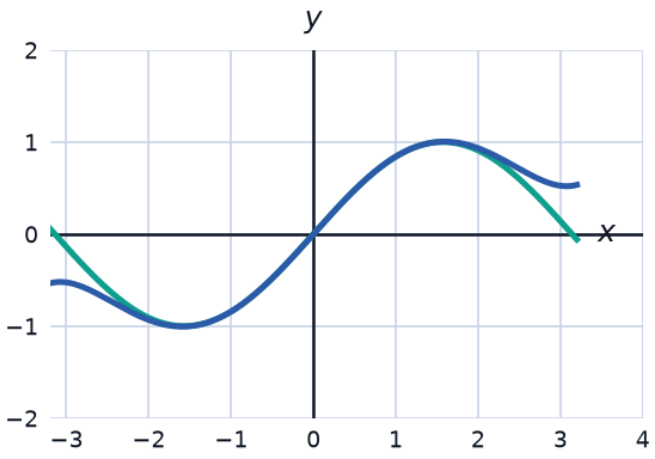
Use the Lagrange Error Bound to estimate the maximum error when approximating $\sin(0.5)$ with the Taylor polynomial $P_3(x) = x - \frac{x^3}{6}$ (degree 3, centered at 0), given $|f^{(4)}(z)| \le 1$ for all z.

9 [Intermediate] • Target: Radius of convergence of Taylor series

State the radius of convergence of the Maclaurin series for e^x, $\sin x$, and $\cos x$.

10 [Intermediate] • Target: Visualizing Taylor polynomial approximations

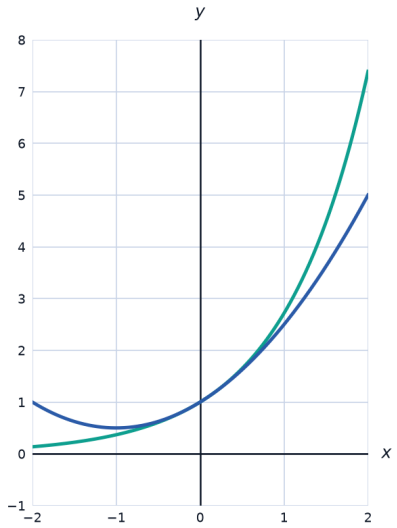
Compare $f(x) = \sin x$ to its degree-5 Maclaurin polynomial $P_5(x) = x - \frac{x^3}{6} + \frac{x^5}{120}$ on $[-\pi, \pi]$, and describe where the approximation is best and worst.



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11 **[Intermediate]** • **Target: Visualizing Taylor polynomial approximations**

Compare $f(x) = e^x$ to its degree-2 Maclaurin polynomial $P_2(x) = 1 + x + \frac{x^2}{2}$ on $[-2, 2]$, and identify visually where P_2 underestimates or overestimates e^x.



12 **[Intermediate]** • **Target: Building Taylor series at a nonzero center**

Find the Taylor series for $f(x) = \cos x$ centered at $a = \frac{\pi}{2}$, up through the cubic term.

13 **[Intermediate]** • **Target: Applying the Lagrange Error Bound to bound accuracy requirements**

Using the degree-4 Maclaurin polynomial for $\cos x$ and the bound $|f^{(5)}(z)| \le 1$, find the maximum error in approximating $\cos(0.3)$.

14 **[Intermediate]** • **Target: The binomial series**

State the binomial series for $(1 + x)^k$ up through the quadratic term, and use it to find the Maclaurin series for $\sqrt{1 + x}$ (i.e. $k = \frac{1}{2}$) up through the quadratic term.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

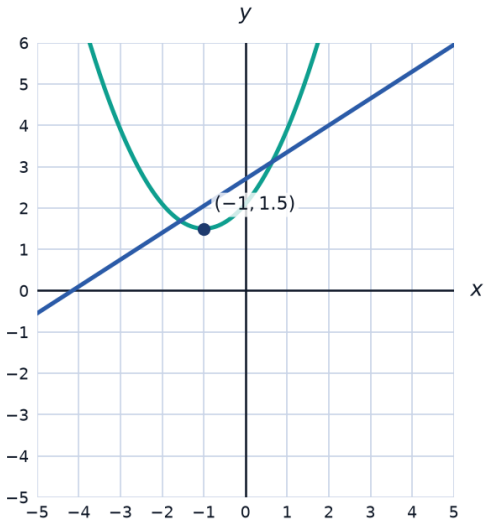
15 **[Challenging]** • **Target: The binomial series** • **Cross-Lesson: 3.02 Vector-Valued Functions**

Use the series found above to approximate $\sqrt{1.1}$ using the first three terms.

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16 [Challenging] • Target: Deriving new Maclaurin series from known ones • Cross-Lesson: 4.01 Polar Curves and Derivatives

Use the known series for e^x to find the Maclaurin series for $\cosh x = \frac{e^x + e^{-x}}{2}$, up through the quartic term.



17 [Challenging] • Target: Building Taylor series at a nonzero center • Cross-Lesson: 4.02 Area in Polar Coordinates

Level 3 · Advanced Let $f(x)$ be a function with $f(2) = 3$, $f'(2) = -1$, $f''(2) = 4$, and $f'''(2) = -12$. Write the third-degree Taylor polynomial $P_3(x)$ for f centered at $x = 2$, and use it to approximate $f(2.1)$.

18 [Challenging] • Target: Applying the Lagrange Error Bound to bound accuracy requirements • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Level 2 · Medium Find the third-degree Maclaurin polynomial $P_3(x)$ for $f(x) = \cos x$.

19 [Challenging] • Target: The binomial series • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Level 3 · Advanced Find the extreme values of $f(x, y) = x^2 + 2y^2$ on the circle constraint $g(x, y) = x^2 + y^2 = 1$ using the Method of Lagrange Multipliers.

20 [Challenging] • Target: Deriving new Maclaurin series from known ones • Cross-Lesson: 6.01 Power Series and Interval of Convergence

(Paper 3 Extended Investigation) Consider the recursive Babylonian sequence defined by $x_0 = 3$ and $x_{n+1} = \frac{1}{2}\left(x_n + \frac{5}{x_n}\right)$ for $n \geq 0$. (a) Use the AM-GM inequality to prove that $x_n \geq \sqrt{5}$ for all $n \geq 1$. (b) Show that the sequence is strictly decreasing for $n \geq 1$, and deduce that $\lim_{n\to\infty} x_n = \sqrt{5}$. (c) Prove the quadratic error relationship $|x_{n+1} - \sqrt{5}| = \frac{(x_n - \sqrt{5})^2}{2x_n}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The Taylor series formula

Use the Maclaurin series for e^u (with $u = 3x^2$) to find the coefficient of x^8 in the Maclaurin series for e^{3x^2}.

a. $\frac{2187}{8}$

b. $\frac{9}{4480}$

c. $\frac{35}{8}$

d. $\frac{27}{8}$

22 [Foundational] • Target: Common Maclaurin series

Using the Lagrange Error Bound $|R_n(x)| \le \frac{M}{(n+1)!}|x-a|^{n+1}$ with $M = 6$, $n = 2$, and $|x-a| = 1$, find the error bound.

a. 2

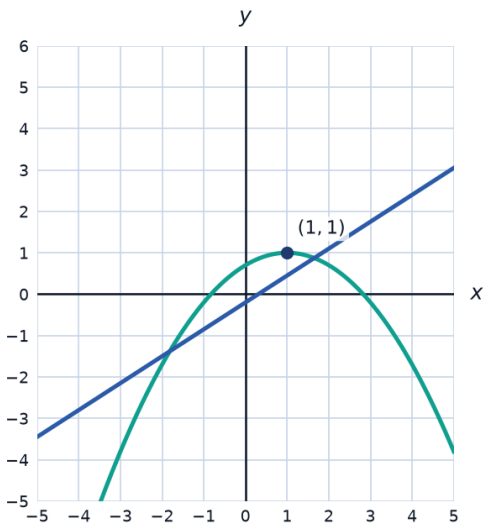
b. $\frac{7}{6}$

c. 3

d. 1

23 [Foundational] • Target: Using known series to build new series

The Maclaurin series for $\sqrt{1+x}$ is a special case of the binomial series with $\alpha =$



a. 2

b. 1

c. $-1/2$

d. $1/2$

24 [Foundational] • Target: The Lagrange error bound

Taylor polynomial $T_2(x)$ for $f(x) = \cos x$ at $x = 0$:

a. $1 - x$

b. $1 - x^2$

c. $1 - x^2/2$

d. $x - x^3/6$

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25 [Foundational] • Target: Radius of convergence of Taylor series

The rate of change of the volume of a spherical balloon being inflated is given by $\frac{dV}{dt} = 12\pi \text{ cm}^3/\text{s}$. If the initial volume of the balloon is $36\pi \text{ cm}^3$, find the volume of the balloon after t seconds. The volume V of a sphere with radius r is $V = \frac{4}{3}\pi r^3$.

a. $V(t) = 12\pi t + 36\pi$

b. $V(t) = \frac{4}{3}\pi t^3 + 36\pi$

c. $V(t) = 4\pi t^3 + 36\pi$

d. $V(t) = 12\pi t^3 + 36\pi$

26 [Foundational] • Target: Visualizing Taylor polynomial approximations

Taylor series of $1/(1 - x)$ at $x = 0$:

a. $\sum x^n/n!$

b. $\sum(-1)^n x^n$

c. $\sum x^{n+1}$

d. $\sum x^n$ for $|x| < 1$

27 [Foundational] • Target: Building Taylor series at a nonzero center

Radius of convergence of $\sum n!\, x^n$:

a. ∞

b. 1

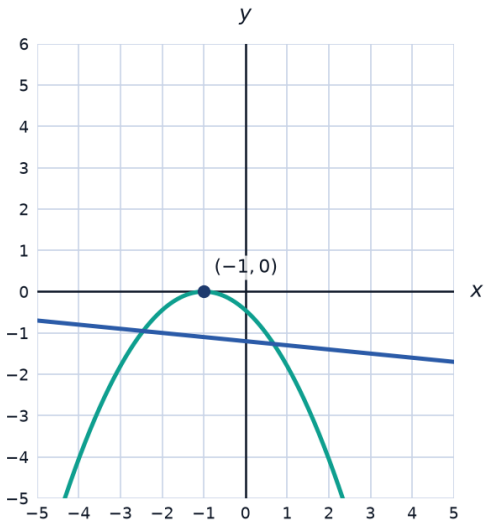
c. 0

d. e

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28 [Foundational] • Target: Applying the Lagrange Error Bound to bound accuracy requirements

Using the first three terms of the Maclaurin series for e^x, approximate $e^{1 \times \frac{1}{4}}$ as a fraction.



a. $\frac{173}{160}$

b. $\frac{41}{32}$

c. $\frac{253}{160}$

d. $\frac{237}{160}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The binomial series

Use the degree-3 Maclaurin polynomial of e^x to approximate its value at $x = \frac{1}{10}$.

a. $\frac{221}{200}$

b. $\frac{6631}{6000}$

c. $\frac{6637}{6000}$

d. $\frac{6031}{6000}$

30 [Intermediate] • Target: Deriving new Maclaurin series from known ones

The Maclaurin series for e^{-x^2} (substitute $-x^2$ for x in e^x series):

a. $\sum \frac{(-1)^n x^{2n}}{n!}$

b. $\sum x^{2n}/n!$

c. $\sum (-x)^n/n!$

d. $\sum x^n/n!$

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31 [Intermediate] • Target: The Taylor series formula

Sum $\sum_{n=1}^{\infty} n(n-1)x^n/(n!)$ simplifies using e^x series to:

a. $e^x(x^2 - x)$

b. xe^x

c. x^2e^x

d. $x^2e^x/2$

32 [Intermediate] • Target: Common Maclaurin series

The power series $\sum a_n x^n$ represents a function that is C^∞ (infinitely differentiable) inside its radius of convergence. This means all derivatives:

a. are polynomials

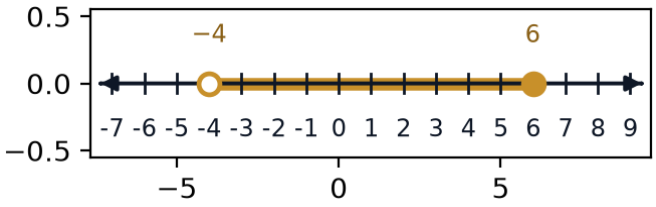
b. are zero

c. exist

d. diverge

33 [Intermediate] • Target: Using known series to build new series

Taylor polynomial error: $|f(x) - T_n(x)| \le \frac{M|x-a|^{n+1}}{(n+1)!}$ where $M =$



a. $\max|f^{(n+1)}|$ on interval

b. $|f^{(n)}(a)|$

c. $n!$

d. $|f(a)|$

34 [Intermediate] • Target: The Lagrange error bound

Taylor series of $f(x) = (1+x)^\alpha$ (binomial series) for $|x| < 1$:

a. $1 + \alpha x$ only

b. $\sum e^{\alpha x}$

c. $\sum x^n/n$

d. $\sum_{n=0}^{\infty} \binom{\alpha}{n} x^n$

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35 [Intermediate] • Target: Radius of convergence of Taylor series

Power series $\sum \frac{(x-2)^n}{3^n}$: centre and radius:

a. centre 0, $R = 3$

b. centre 3, $R = 2$

c. centre 2, $R = 1/3$

d. centre 2, $R = 3$

36 [Intermediate] • Target: Visualizing Taylor polynomial approximations

Using power series, compute $\lim_{x \to 0} \frac{1 - \cos x}{x^2}$:

a. 1/2

b. 1

c. 0

d. 2

37 [Intermediate] • Target: Building Taylor series at a nonzero center

Power series $\sum_{n=0}^{\infty} n(x-1)^n$. Radius of convergence:

a. 0

b. 1

c. 2

d. ∞

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Applying the Lagrange Error Bound to bound accuracy requirements • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Taylor remainder (Lagrange form): $R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$ for some c between a and x. This bounds the:

a. average error

b. maximum error

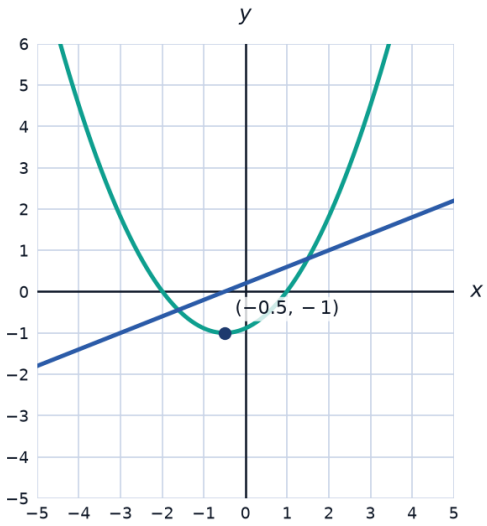
c. minimum error

d. exact error

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39 [Challenging] • Target: The binomial series • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Hypergeometric series ${}_2F_1(a,b;c;x)=\sum_{n=0}^{\infty}\frac{(a)_n(b)_n}{(c)_n n!}x^n$ (Pochhammer symbol). Radius of convergence:



a. depends on a, b, c

b. always ∞

c. always 1

d. always 0

40 [Challenging] • Target: Deriving new Maclaurin series from known ones • Cross-Lesson: 6.01 Power Series and Interval of Convergence

If a function equals its Taylor series everywhere, it is called:

a. analytic

b. polynomial

c. differentiable

d. continuous

41 [Challenging] • Target: The Taylor series formula • Cross-Lesson: 1.01 Integration by Parts

Maclaurin series for e^x: first four terms:

a. $1+x+x^2+x^3$

b. $1+x+x^2/2!+x^3/3!$

c. $x+x^2+x^3$

d. $1-x+x^2-x^3$

42 [Challenging] • Target: Common Maclaurin series • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

$\lim_{x\to 0}\frac{1-\cos x}{x^2}$. Using L'Hôpital twice or Taylor series:

a. 1/2

b. 0

c. 1

d. $-1/2$

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43 [Challenging] • Target: Using known series to build new series • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Bessel function $J_0(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(m!)^2}\left(\frac{x}{2}\right)^{2m}$. Radius of convergence:

a. 1

b. ∞

c. π

d. 2

44 [Challenging] • Target: The Lagrange error bound • Cross-Lesson: 3.01 Parametric Equations

The Padé approximant $[m/n]$ for $f(x)$ is a rational function $P_m(x)/Q_n(x)$ matching the Taylor series to order $m+n$. It typically converges more:

a. slowly than Taylor

b. only at 0

c. rapidly than Taylor for poles

d. never

45 [Challenging] • Target: Radius of convergence of Taylor series • Cross-Lesson: 3.02 Vector-Valued Functions

The binomial series $(1+x)^{1/2}$ up to x^2:

a. $1 + x - x^2$

b. $1 + x/2 + x^2/8$

c. $1 + x/2$

d. $1 + x/2 - x^2/8$

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